



OLA RIDE BOOKING ANALYTICS PROJECT

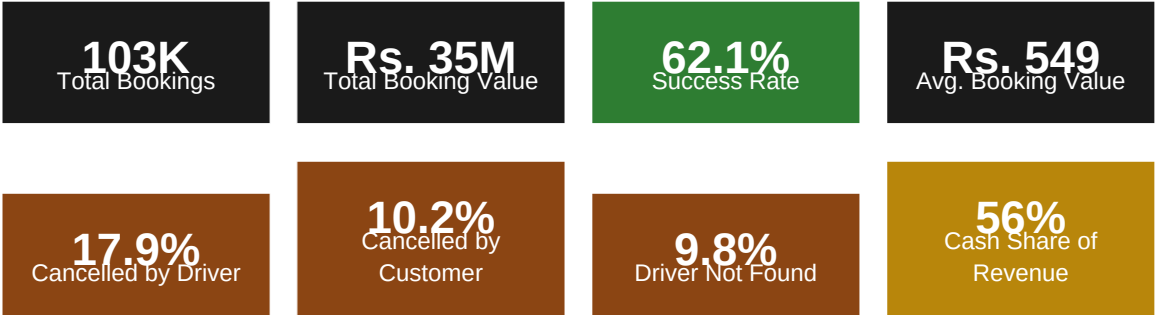
SQL Data Modeling | Power BI Dashboard | Business Insights

Bengaluru City Ride-Hailing Dataset • July 2024 • 1,03,025 Bookings
Prepared as a Data Analyst Portfolio Project

Executive Summary

This project analyzes one month (July 2024) of ride-booking data for OLA's Bengaluru operations, covering 1,03,025 bookings across 7 vehicle categories. The dataset was structured and queried using SQL (database design, views, aggregations), and visualized in a 5-page interactive Power BI dashboard covering overall performance, vehicle-type economics, revenue, cancellations, and ratings.

Top-Line KPIs



Key Business Insights

- Only **62%** of bookings complete successfully — a combined **38%** are lost to driver cancellations, customer cancellations, or 'driver not found' events, representing the single biggest revenue leak in the funnel.
- **Driver-side cancellations (17.9%)** outweigh customer-side cancellations (10.2%) by nearly 2x, pointing to supply-side issues (driver availability/behavior) as the primary cancellation driver.
- Booking value is spread almost evenly across all 7 vehicle types (Rs. 7.9M–Rs. 8.2M each), meaning no single vehicle category dominates revenue — OLA's Bengaluru demand is well diversified.
- **Cash (Rs. 19M)** and **UPI (Rs. 14M)** together make up ~96% of revenue; card payments are negligible, signalling a cash/UPI-first customer base.
- Driver and customer satisfaction are both consistently high and nearly symmetric (~3.98–4.01 / 5) across every vehicle type, indicating a stable service experience with no underperforming category.
- Ride volume and distance both show a clear **end-of-month spike** (day ~30), suggesting weekend/month-end demand surges worth targeting with dynamic pricing or driver incentives.

Dashboard 1 — Overall Performance

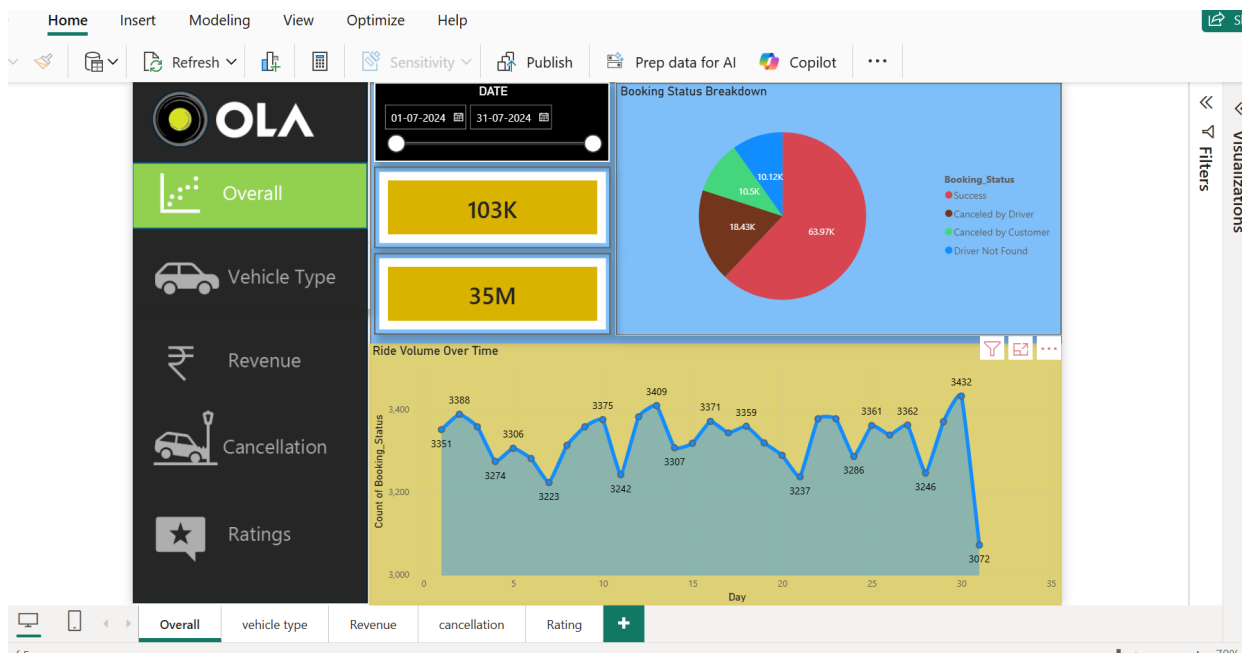


Fig 1. Overall KPIs, booking status breakdown, and daily ride volume trend.

Insights

- Of 103K total bookings, 63.97K (62.1%) were successful — the baseline conversion rate for the platform.
- Cancellations split as: 18.43K (17.9%) by driver, 10.5K (10.2%) by customer, and 10.12K (9.8%) as 'driver not found'.
- Daily ride volume is stable in the 3,200–3,400 range on most days, with a sharp dip on the final day of the month (month-end data cutoff) and a peak around day 9–10 and day 30.

Dashboard 2 — Vehicle Type Analysis

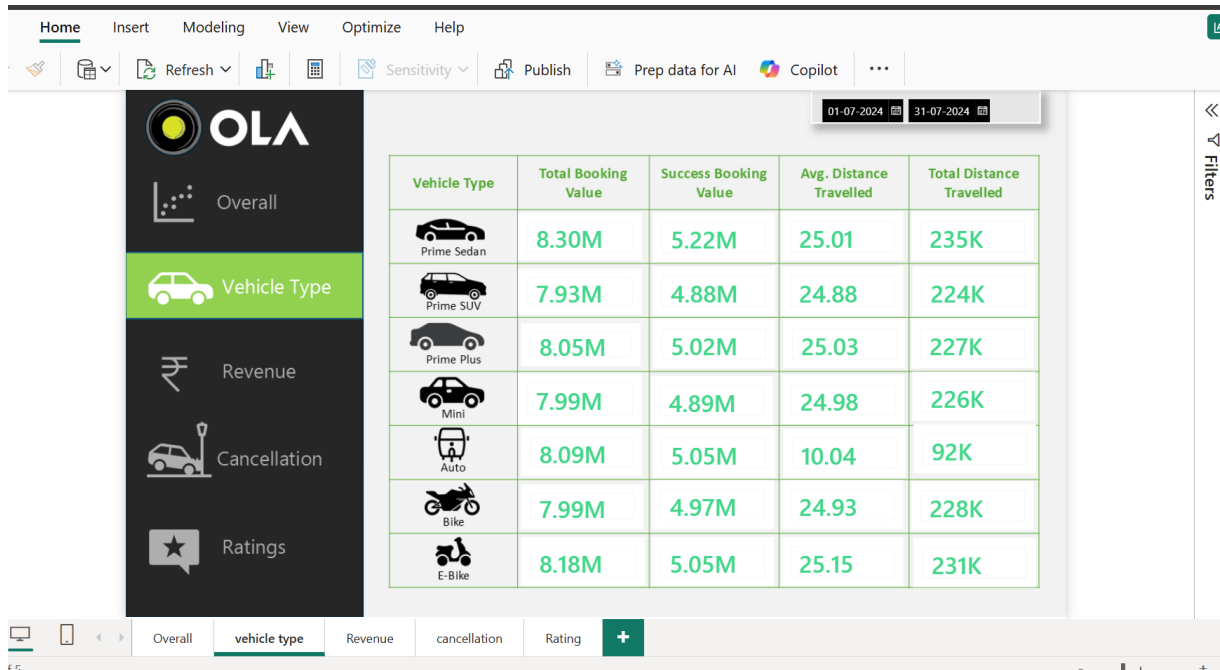


Fig 2. Booking value, success value, average distance, and total distance by vehicle type.

Insights

- E-Bike (Rs. 8.18M) and Auto (Rs. 8.09M) generate the highest total booking value; Prime SUV (Rs. 7.93M) the lowest — the spread across categories is under 3%, showing balanced demand.
- Auto has a distinctly lower average ride distance (10.04 km) than every other category (~25 km), consistent with autos being used for short, intra-locality trips.
- Despite the short distance, Auto still ranks among the top vehicle types by revenue — indicating high trip frequency rather than high per-trip value drives its revenue.

Dashboard 3 — Revenue Analysis

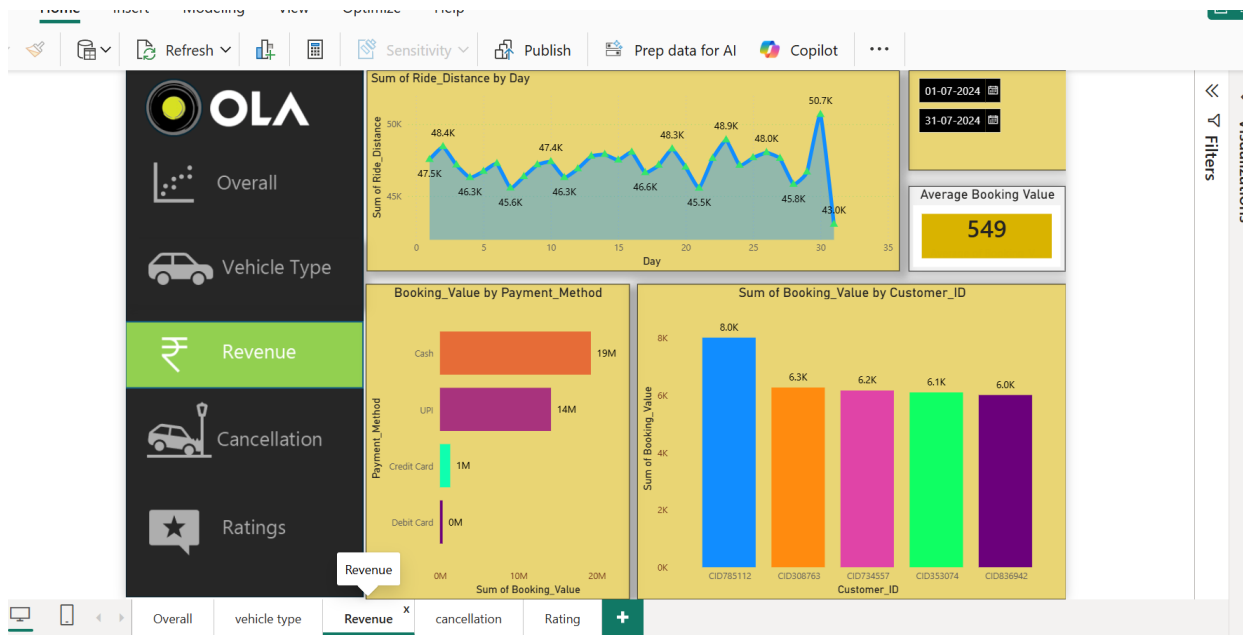


Fig 3. Ride distance trend, revenue by payment method, and top 5 customers by booking value.

Insights

- Cash (Rs. 19M) is the leading payment method, followed by UPI (Rs. 14M); credit and debit cards combined contribute under Rs. 1M — digital card payments are a clear growth opportunity.
- The top 5 customers each contributed Rs. 6K–Rs. 8K in booking value, led by customer CID785112 at Rs. 8.0K.
- Average booking value sits at Rs. 549, with daily ride-distance totals climbing toward month-end, peaking at 50.7K km on day 30 — mirroring the ride-volume spike seen in the Overall dashboard.

Dashboard 4 — Cancellation Analysis

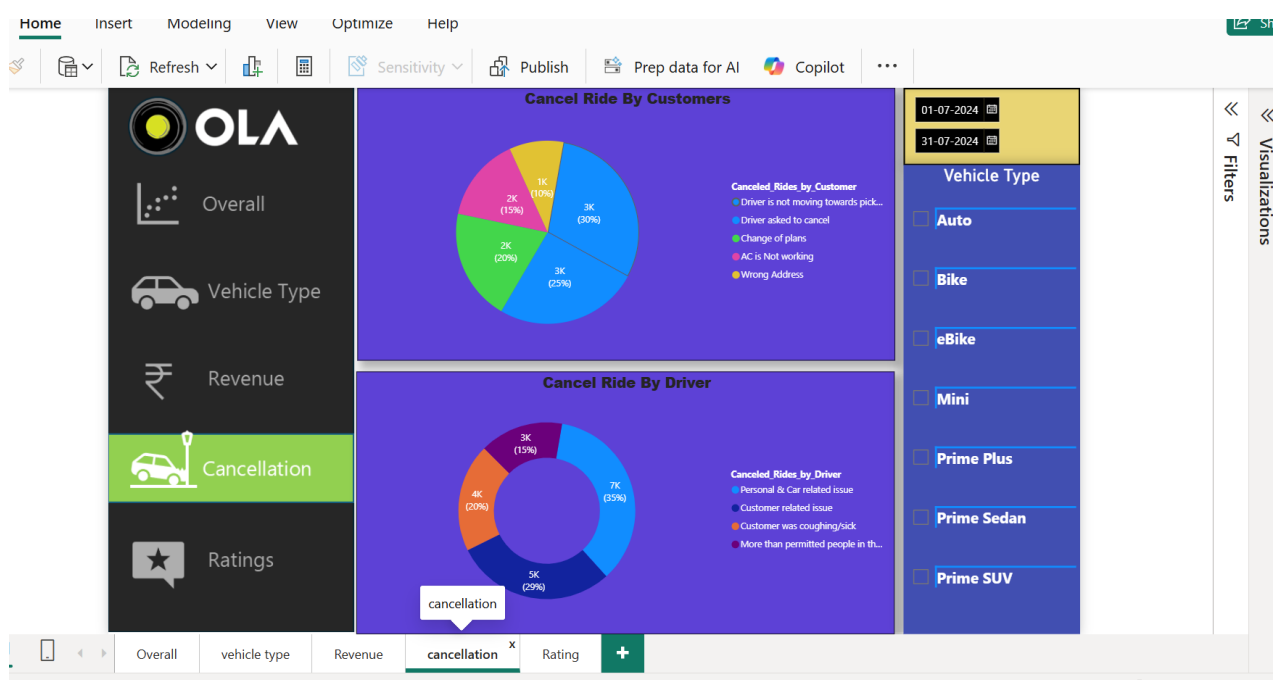


Fig 4. Cancellation reasons split by customer-initiated vs driver-initiated cancellations.

Insights

- Customer-side cancellations are fairly evenly spread across five reasons, with the largest single reason accounting for ~30% and the smallest for ~10% — no one factor dominates on the customer side.
- Driver-side cancellations are far more concentrated: 'Personal & car related issues' alone accounts for ~35%, and together with 'Customer related issue' (~29%) these two reasons make up nearly two-thirds of all driver cancellations.
- Because driver cancellations (17.9%) are the largest loss category overall, tackling 'personal & car related issues' specifically would have the single biggest impact on improving the platform's completion rate.

Dashboard 5 — Ratings Analysis

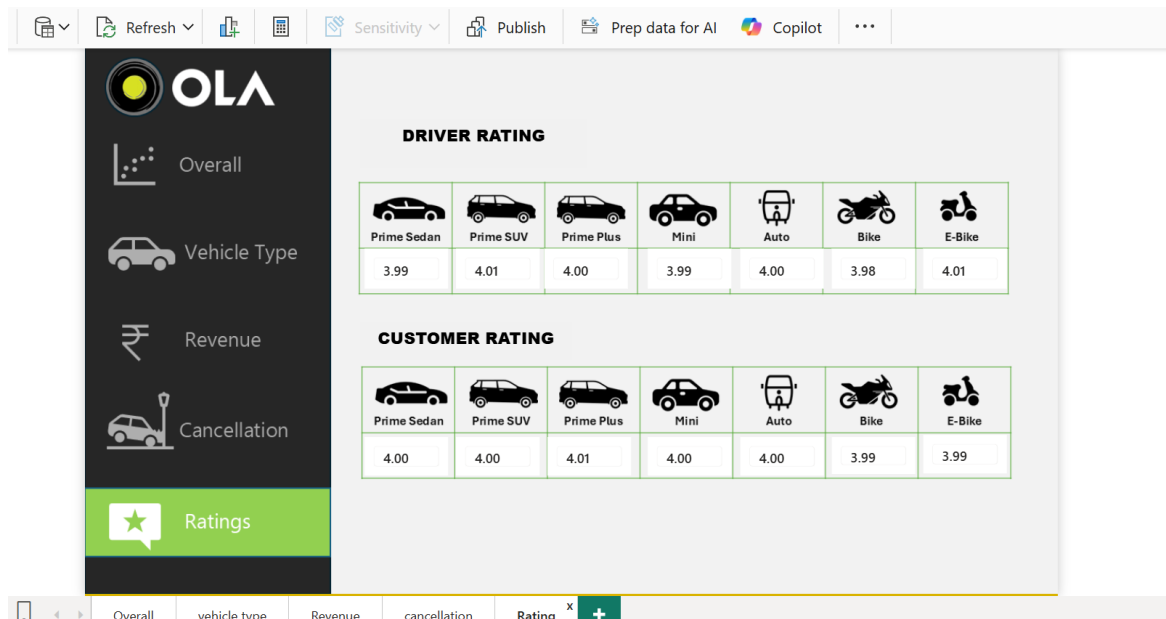


Fig 5. Average driver and customer ratings across all 7 vehicle types.

Insights

- Driver ratings range narrowly from 3.98 (Bike) to 4.01 (Prime SUV, E-Bike) — a spread of just 0.03, showing very consistent driver-side service quality across categories.
- Customer ratings are even tighter, from 3.99 to 4.01 across all vehicle types, suggesting drivers rate customers uniformly well regardless of vehicle category.
- The near-identical driver and customer scores (both hovering ~4.0/5) indicate a healthy, balanced two-sided marketplace with no friction concentrated in any one vehicle segment.

SQL Analysis Layer

Alongside the Power BI dashboards, the raw bookings data was loaded into a relational database (**Ola.Bookings**) and queried with 10 SQL views answering core business questions:

1. Retrieve all successful bookings (Successful_Bookings view)
2. Average ride distance per vehicle type (AVG_Distance_By_Vehicle)
3. Total rides cancelled by customers (Cancelled_Rides_By_Customer)
4. Top 5 customers by ride count (Top_5_Customers)
5. Driver cancellations due to personal & car issues (Cancelled_By_Driver_Personal_Car)
6. Max/min driver ratings for Prime Sedan (Prime_Sedan_Rating_Range)
7. All UPI-paid rides (UPI_Payments)
8. Average customer rating per vehicle type (AVG_Customer_Rating_By_Vehicle)
9. Total booking value of successful rides (Total_Successful_Booking_Value)
10. Incomplete rides with reasons (Incomplete_Rides_List)

Full schema, import script, and all 10 queries are available in the accompanying **OLA_Analysis_Structured.sql** file in the project repository.

Recommendations

- Investigate driver-side 'personal & car related' cancellations — the largest single loss driver — through targeted driver support, vehicle-maintenance partnerships, or better trip-matching.
- Push UPI/card incentives to reduce cash dependency (currently 56% of revenue), improving payment reconciliation.
- Use the day-30 demand spike pattern to test dynamic pricing or surge driver incentives around weekends/month-end.
- Since ratings are uniformly high across vehicle types, focus retention efforts on completion rate (the 38% drop-off) rather than service quality, which is not the bottleneck.

Tools used: SQL (PostgreSQL) · Power BI · Excel/CSV data source · Bengaluru ride-booking dataset (July 2024)